

Artifacts From the Zoom Coil and the Ways to Reduce/Eliminate Them

Origin of Artifacts

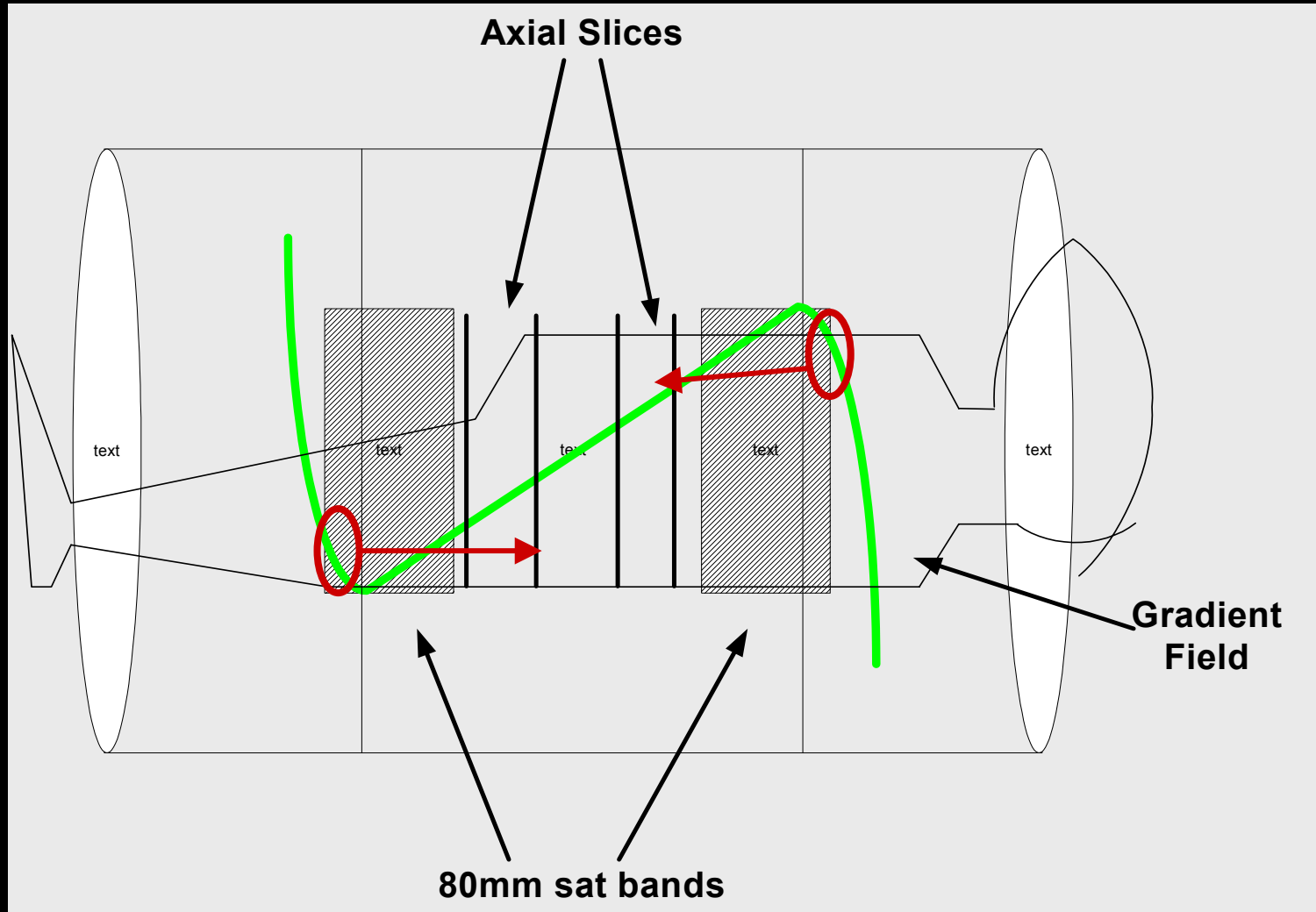
The shorter length of the TwinSpeed Zoom Gradient coil can potentially cause image artifacts to be introduced with “routine” protocols. We will discuss the issues, causes, and fixes for these artifacts.

As illustrated by the diagram on the next slide, tissue excited outside the limit of the Zoom coil can “wrap” back into the phase FOV and appear as a streak or zipper.

In almost all cases, using the Whole Gradient will eliminate this type of artifact, but for the Zoom Gradient, remedies include proper coil selection and/or SAT pulse modifications.

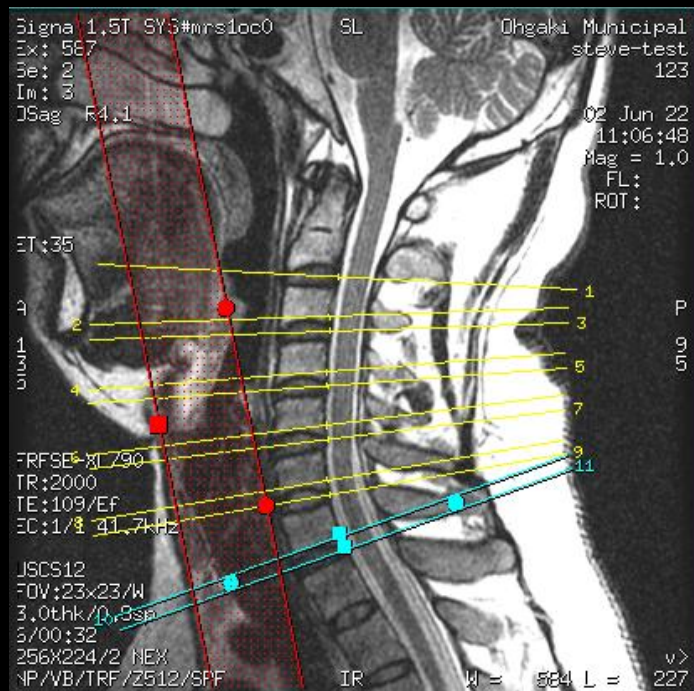
Global MR

Origin of Artifacts



Global MR

Zoom Artifacts – C Spine



Angled Anterior SAT pulse



Zipper artifact

- Anterior Spatial SAT excites tissue beyond the Zoom coil - Crusher gradient from Zoom coil becomes in-effective.
- 3-Coil CTL (CS123) Zoom Mode receives signal from tissue excited by the SAT pulse outside the FOV, and wraps into the phase direction.

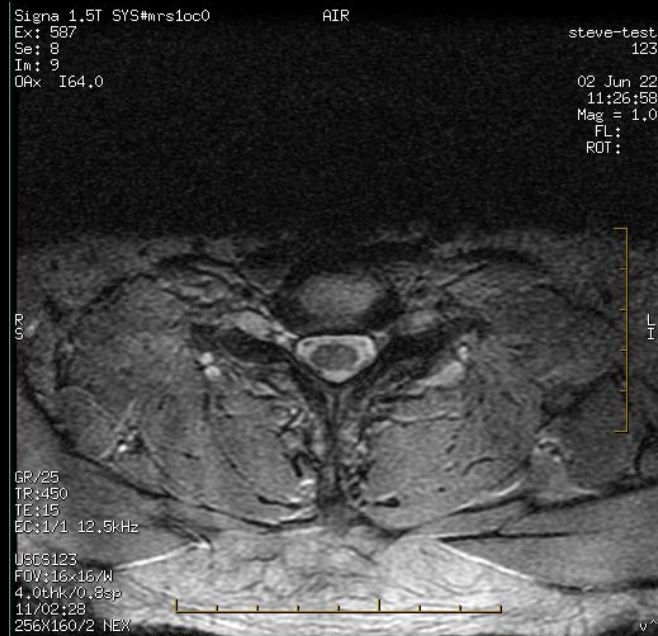
Solution - Use 2-Coil CTL (CS12) effectively eliminates the artifacts.

Global MR

Workaround:
Whole Body mode,

or

Solution:
Use Zoom with 2 coil
elements

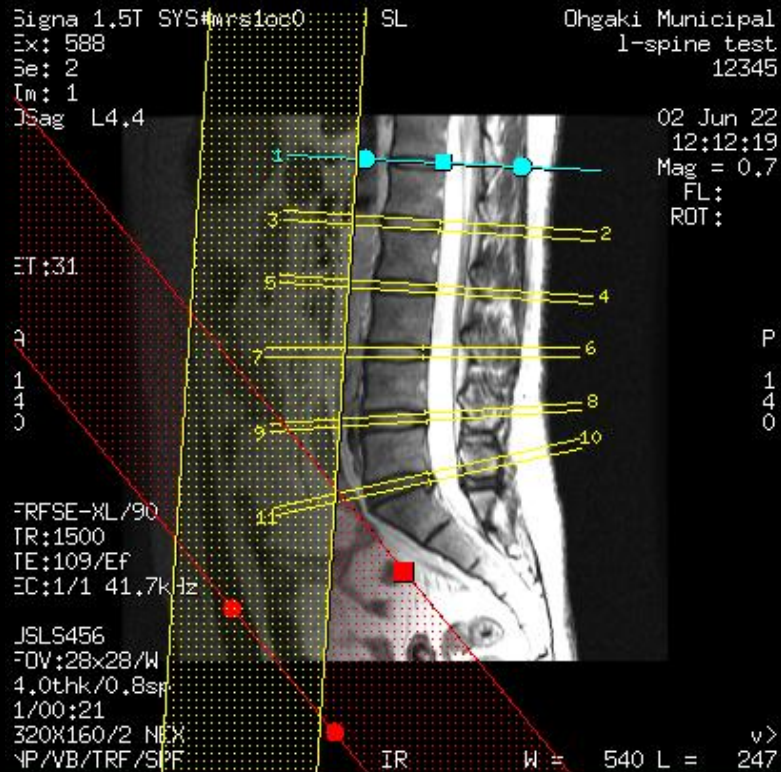


3-Coil CTL (CS123)
Clean Whole Body mode.

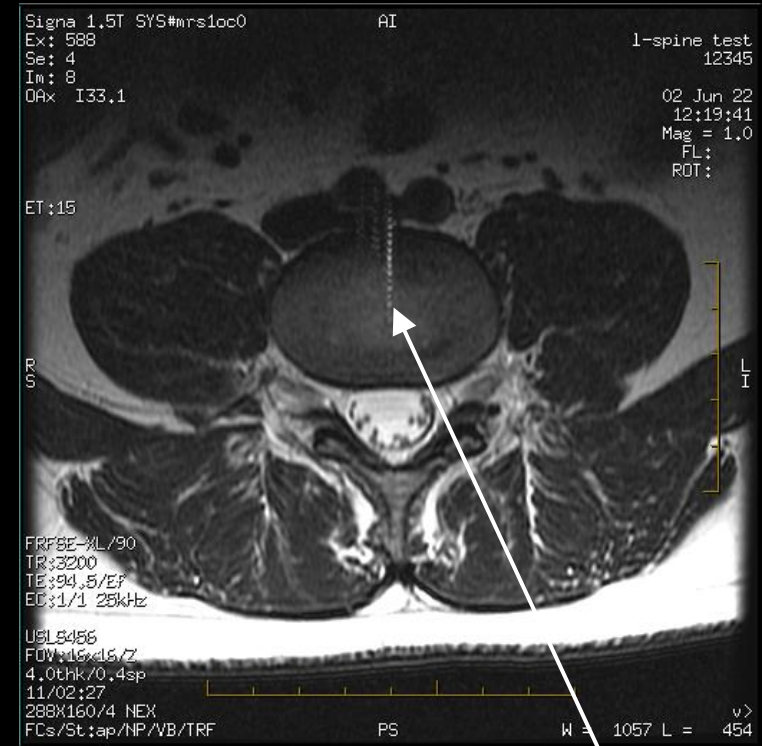


2-Coil CTL (CS12)
Clean Zoom mode.

Zoom Artifacts – L Spine



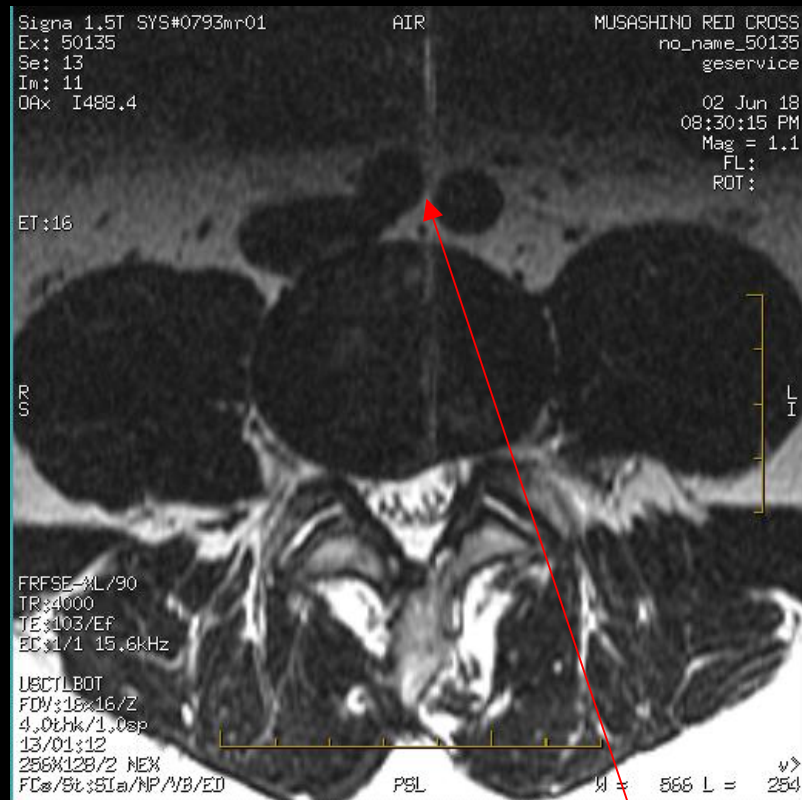
Angled Anterior + Posterior 80mm SAT pulses



Zipper artifact

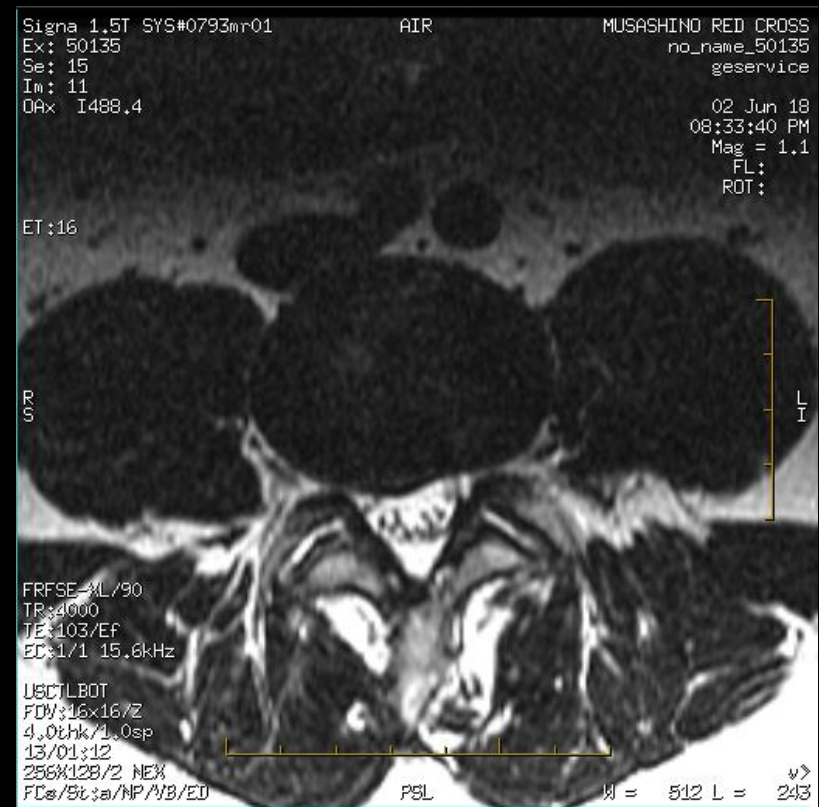
- Spatial SAT excite tissue beyond the Zoom coil - Crusher gradient from the Zoom becomes ineffective.
- 3-Coil CTL (LS456) Zoom Mode can receive signal from tissue excited by the wide SAT pulse outside the FOV, and wrap into the phase direction.

Zoom Artifacts – L Spine



Zipper-like artifact with SI sat pulses applied.

Solution

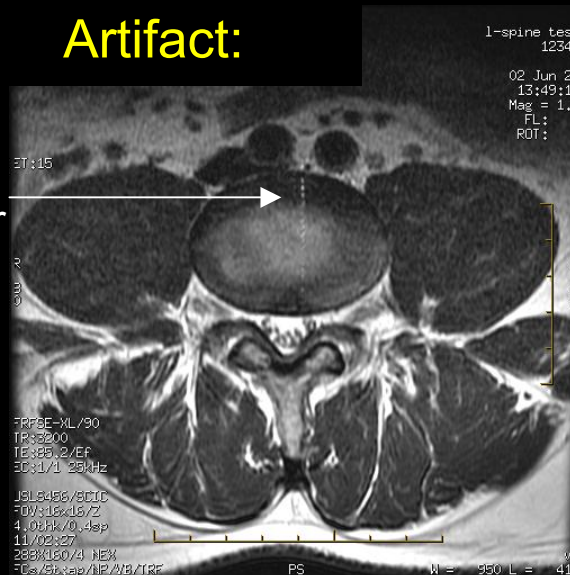


Artifact is gone when SI sat pulses were removed.

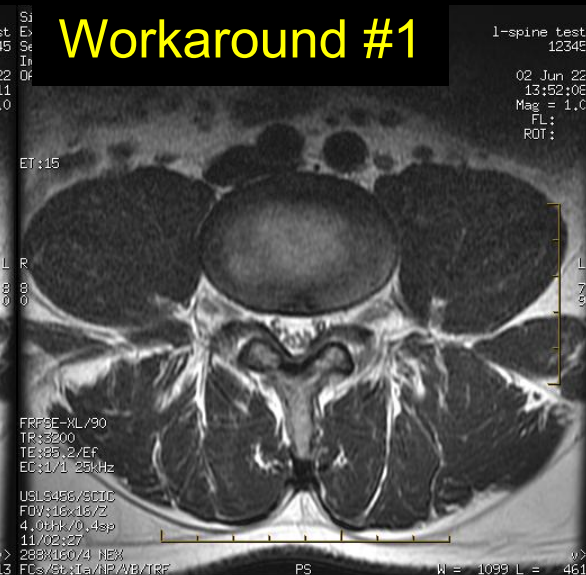
Global MR

Artifact:

Angled
Anterior+Posterior
SAT pulse 80mm
thick



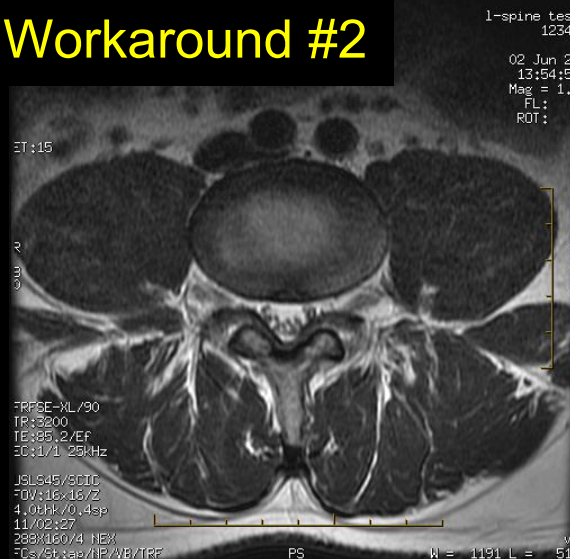
Workaround #1



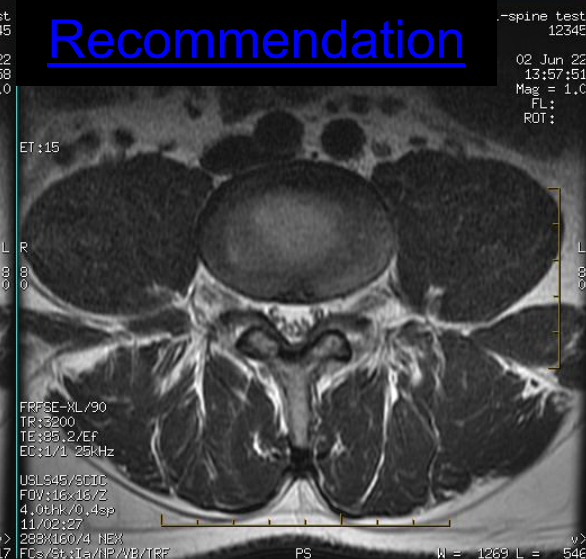
3-Coil CTL
(LS456) with
30mm SAT

Workaround #2

2-Coil CTL
(LS45) with
80mm SAT



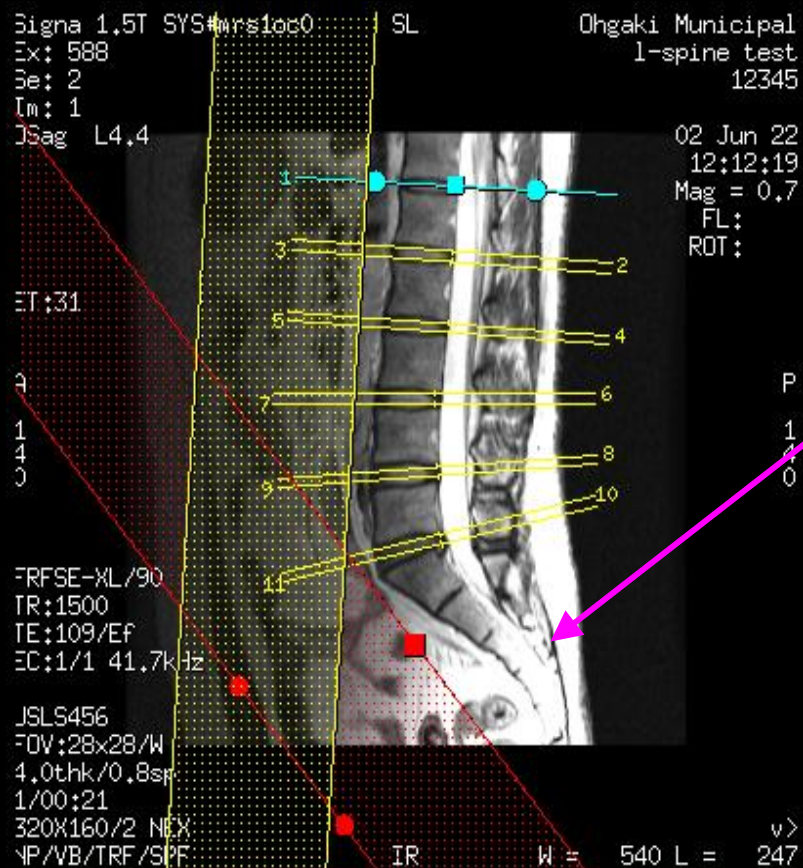
Recommendation



2-Coil CTL
(LS45) with
30mm SAT

Global MR

L Spine – Special consideration for Asia



For Lumbar Spine, it is most common to use the lower 3 CTL coils(LS456).

However, due to the shorter average height of the Asian population, the area of coil #6 is often outside the scan FOV, causing the Zoom coil artifact.

When scanning in Zoom Mode, tissue under coil #6 is excited by the **wide** SAT pulse outside the scan FOV, and can wrap into the phase direction causing the artifact. For a taller patient, it might be coil #4 that is outside the FOV.

A potential recommended scan procedure could include a change to the coils used for the initial localizer scan, described in the next slide.....

Global MR

Method



Origin of Artifacts

In addition to selecting only 2 coils for Zoom Mode scanning to reduce artifacts, there is a SNR gain when using 2 coils vs. 3 coils. Since the 3rd coil is typically outside the scan FOV, there is little to no signal received; however, there is noise added to the sum of the 3 coils. This reduces the overall SNR. See the chart on the next slide comparing SNR of CS123 vs. CS12.

Global MR

Improved SNR with 2 Coil CTL

Measured 22 slices with a phantom:

Average SNR

3 CTL elements CS123 – 17.6

2 CTL elements CS12 – 22.4

27% Increase in SNR !

CS123			CS12		
Signal	Noise	SNR	Signal	Noise	SNR
636.30	37.19	17.11	617.05	28.61	21.57
638.42	36.99	17.26	622.37	28.34	21.96
648.39	37.78	17.16	624.68	28.92	21.60
650.08	37.68	17.25	631.11	28.02	22.52
653.76	37.64	17.37	635.04	27.69	22.93
654.16	36.82	17.77	636.06	28.72	22.15
664.84	37.48	17.74	638.33	27.91	22.87
661.65	36.84	17.96	639.60	29.01	22.05
664.17	37.50	17.71	641.50	28.30	22.67
663.59	36.79	18.04	641.60	28.27	22.70
671.44	37.24	18.03	645.50	28.43	22.70
664.59	37.76	17.60	643.53	28.95	22.23
667.17	37.51	17.79	641.91	28.11	22.84
665.72	36.52	18.23	643.99	28.65	22.48
670.88	37.18	18.04	643.79	28.85	22.32
662.73	37.65	17.60	642.62	28.30	22.71
665.41	37.28	17.85	641.02	28.29	22.66
662.84	37.73	17.57	640.37	27.97	22.89
664.72	37.43	17.76	639.31	28.84	22.17
658.31	37.57	17.52	636.74	28.76	22.14
658.88	38.02	17.33	633.36	28.77	22.01
655.04	37.15	17.63	633.48	28.67	22.10
	Average =	17.65		Average =	22.38
	STD =	0.31		STD =	0.41

Reduction of Artifacts

Review:

- Zoom Mode can introduce artifacts by introduction of signal received outside the Zoom Gradient limit....less of an issue with Whole Gradient mode
- Eliminating or narrowing the SAT pulse width can reduce the artifact.
- Selecting the proper coils (inside Zoom limit) can reduce the artifact.
- 2-coil CTL has ~27% more SNR than 3-coils, when scan FOV allows.

Abdomen Artifacts

Other issues and artifacts observed include Abdominal Imaging.....

- Ghosting when using 2 RxR Resp Triggering:

Try 1 or 3 RxR and increase the breath per minute
(seems to be more accurate at sensing the next R wave)

- Irregular FatSat:

Use High Order Shim (if available)

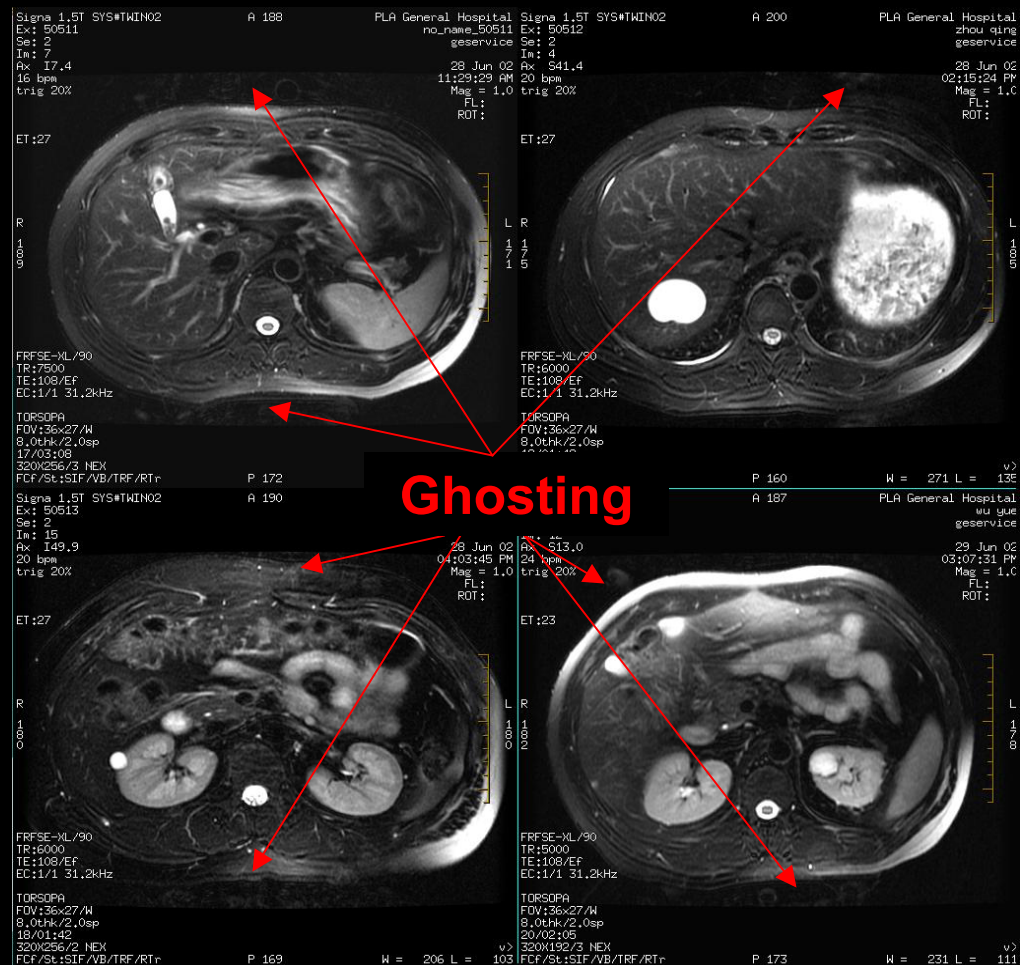
- “Dot” Artifact:

Use Phase Coorection for FSE, “mempx” for SE T1

Global MR

Specific IQ Issues - Ghosting

Areas of Ghosting despite proper set-up of Resp. Triggering:



Ghosting – Suggested Improvement

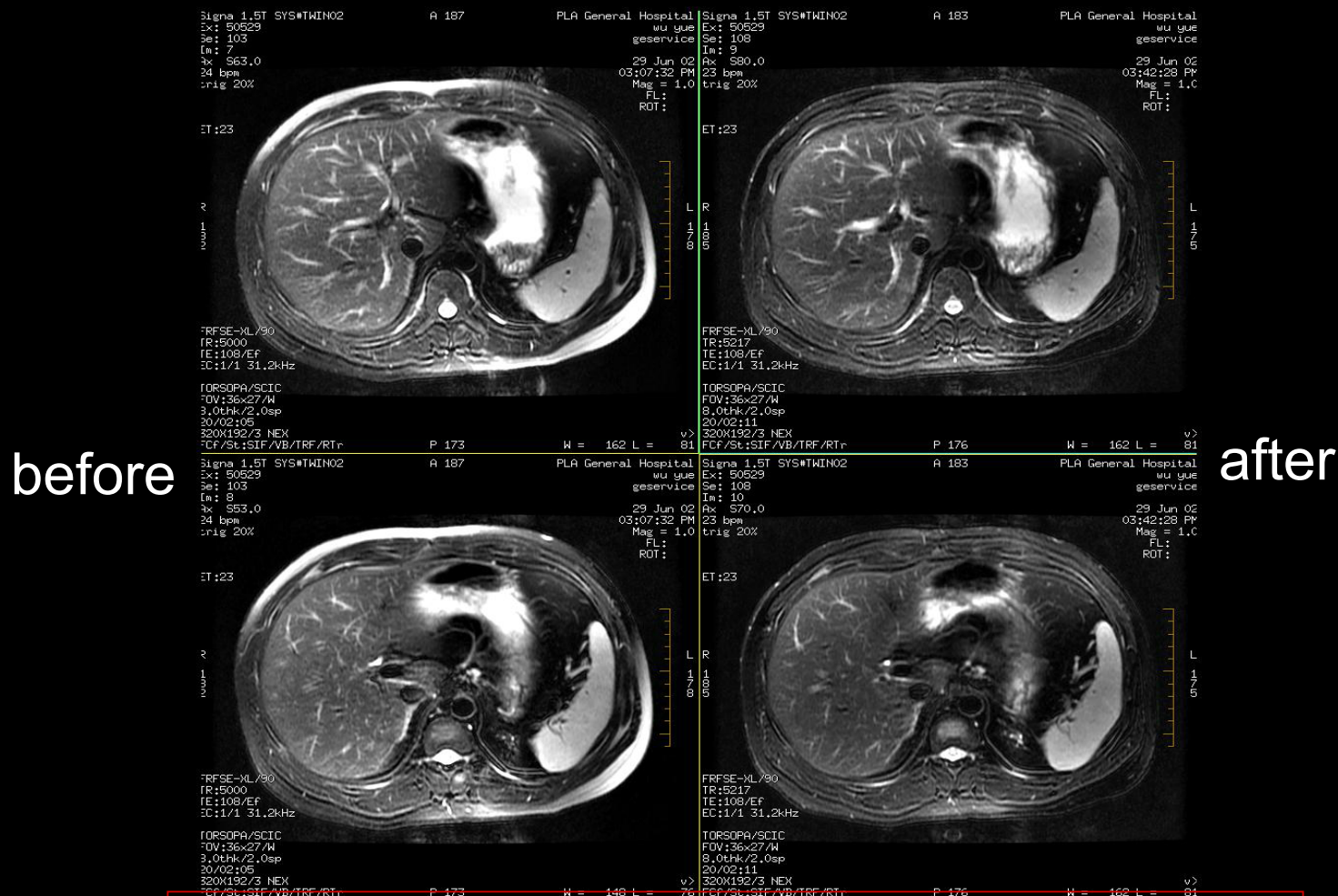
Ghosting

Reduced Ghosting

1 RxR AND
3 RxR (scan ok)

FatSat – Suggested Improvement

FatSat Comparison before / after High Order Shim:



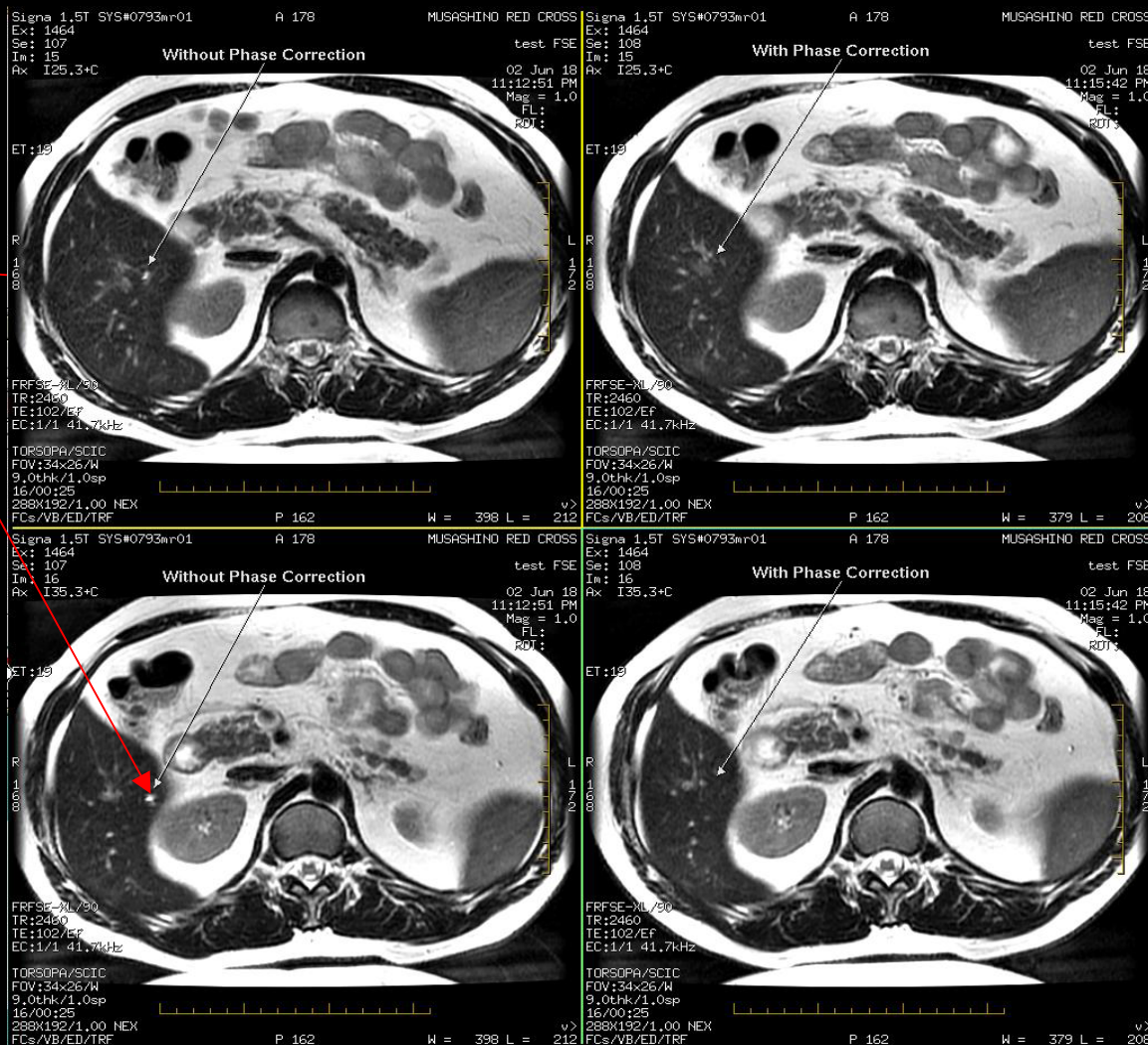
HOS – Looks promising, need more clinical data

Specific IQ Issues – “Dot Artifact”

“Dot” Artifact removed using Phase Correction:

“Dot”
Artifact

Without
Phase
Correction



Solution

With Phase
Correction